



Conference Abstract

Quantifying groundwater recharge in karstic mediterranean environments controlling surface water and energy transfers

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Abstract

In Mediterranean environments, droughts are identified as a major factor of vulnerability, in particular in forestry ecosystems which are exposed to increasingly frequent and intense droughts induced by climatic changes. Moreover, these ecosystems are mainly located in karstic environments, with a crucial importance of groundwater for anthropic and vegetation uses, but also with complex and heterogeneous surface and hydrogeological processes. For instance, recent ecophysiological and isotopic studies have shown that tree roots are able to extract water deep enough in the epikarst to sustain transpiration during water stress periods (Carrière et al. 2020).

However, the quantification of water stocks, aquifer recharge and their dynamics are not yet fully established in such a context. This calls for using models suited to the complexity of the environment, able to improve knowledge of both groundwater recharge and forests

hydic processes. The modelling challenges are to adapt models to karstic environment constraints, in order to:

1. jointly simulate diffuse infiltration into the superficial part of the root zone and fast preferential infiltration into the network of karstic fractures, and
2. simulate the transpiratory and water extraction processes throughout the root zone.

Toward this objective, we improved a detailed SVAT model (SiSPAT, Braud et al. 1995) devoted to simulate energy and water exchanges into the epikarst, by including a new groundwater module able to allocate surface runoff into the network of fractures. This new version was implemented, and then evaluated for the first time on two mediterranean forest sites included in the ICOS network, namely the forest sites of Font-Blanche (Bouches-du-Rhône), managed by URFM INRAE and URFM 2025) and Puéchabon (Hérault), managed by CEFE (CNRS and CEFE 2025). Performances obtained between observed and modeled soil water content, soil temperature and energy fluxes were particularly good. These results highlighted the necessity of representing both diffuse and preferential flows in SVAT modelling for karstic areas, to correctly reproduce surface flux dynamics. Moreover, it was also shown that preferential infiltration builds up deep water storage throughout the year, and considerably improves transpiration processes during water stress periods. This newly integrated process also significantly affected the simulation of the other hydrological balance components, largely reducing runoff and water storage into the soil, through introducing a groundwater recharge flow as observed in this area. Finally, results allowed us to reduce uncertainties in quantitative estimates of groundwater recharge, through a better control of evapotranspiration at the surface.

Our modelling approach shows new promising results, both in terms of performances obtained on two karstic forest sites and of the knowledge of processes at the surface-groundwater interface. This study opens further perspectives on the integrated functioning of the critical zone in karstic environments from mechanistic models. Next steps will concern the coupling between water and carbon cycles, through the development of a new module devoted to photosynthesis and improved stomatal conductance. This step is under progress, and will allow us to study the impacts of future droughts in a context of climatic changes.

Keywords

karst, forest, mediterranean climate, drought, modelling, energy, hydrology, evapotranspiration, water stress

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Conflicts of interest

The authors have declared that no competing interests exist.

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